

TEK4090

Introduction to Modern Control

Lecture 5: Estimation

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Recap from Last Time

- ▶ Pole placement

Recap from Last Time

- ▶ Pole placement
- ▶ Optimal control

Recap from Last Time

- ▶ Pole placement
- ▶ Optimal control
- ▶ Linear Quadratic Regulator

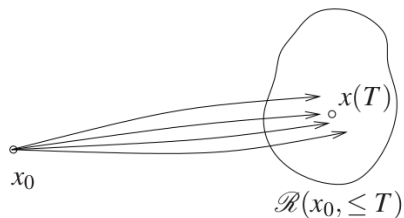
Recap from Last Time

- ▶ Pole placement
- ▶ Optimal control
- ▶ Linear Quadratic Regulator
- ▶ MPC and Building Control

Reading Assignment

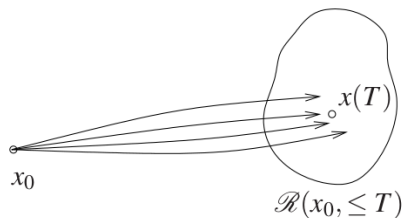
- ▶ **Åstrom & Murray:** Ch. 7
- ▶ **Crassidis & Junkins:**
 - ▶ Kalman filters: Ch. 3.1–3.3
 - ▶ Spacecraft Dynamics: Ch. 6.1, Appendix A.7-A.8
 - ▶ Probability: Appendix C

Reachability



The **reachable set** $\mathcal{R}(x_0, T)$ is the set of all points x_f such that there exists an input $u(t)$ that steers the system from $x(0) = x_0$ to $x(T) = x_f$.

Reachability



The **reachable set** $\mathcal{R}(x_0, T)$ is the set of all points x_f such that there exists an input $u(t)$ that steers the system from $x(0) = x_0$ to $x(T) = x_f$.

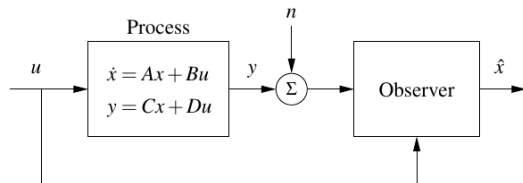
A system is **reachable** if for any $x_0, x_f \in \mathbb{R}^n$, there exists $T > 0$ and $u : [0, T] \mapsto \mathbb{R}$ such that $x(0) = x_0$ and $x(T) = x_f$.

Observability

Feedback requires a measurement of the system.

How do we use measurements of a system in the control loop?

► Consider the LTI system (A, B, C, D)

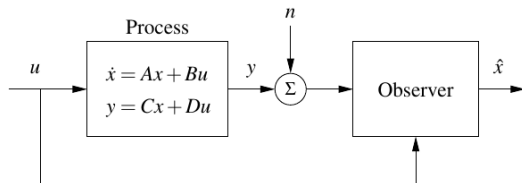


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How do we use measurements of a system in the control loop?

- ▶ Consider the LTI system (A, B, C, D)
- ▶ Suppose y is corrupted by noise n

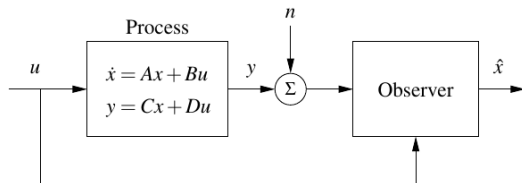


Observability

Feedback requires a measurement of the system.

How do we use measurements of a system in the control loop?

- ▶ Consider the LTI system (A, B, C, D)
- ▶ Suppose y is corrupted by noise n
- ▶ We can construct an **observer** to produce the estimate $\hat{x}$



Observability

Challenges/Questions:

1. How 'rich' does our measurement need to be in order to be useful?

Observability

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2. How do you design an observer?

Observability

Challenges/Questions:

1. How 'rich' does our measurement need to be in order to be useful?
2. How do you design an observer?
3. How does the interconnection between the observer and the controller work?

Observability

Answer to **Challenge 1**:

Observability

Answer to **Challenge 1**:

Definition:

A system $\Sigma := (A, B, C, D)$ is said to be **observable** if, for any $T > 0$, it is possible to reconstruct the state $x(T)$ from measurements of $y(t)$ and $u(t)$ on $[0, T]$.

Observability

Answer to **Challenge 1**:

Definition:

A system $\Sigma := (A, B, C, D)$ is said to be **observable** if, for any $T > 0$, it is possible to reconstruct the state $x(T)$ from measurements of $y(t)$ and $u(t)$ on $[0, T]$.

So, when is a system observable?

Observability

It's useful just to consider the autonomous system

$$\dot{x} = Ax$$

$$y = Cx.$$

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$$y = Cx.$$

Immediately note: if C is invertible, then we can recover $x(t)$ from $x(t) = C^{-1}y(t)$.

Observability

It's useful just to consider the autonomous system

$$\dot{x} = Ax$$

$$y = Cx.$$

Immediately note: if C is invertible, then we can recover $x(t)$ from $x(t) = C^{-1}y(t)$.

In general, C is rank-deficient and can have fewer rows than x .

Observability

neglect the input: $\dot{x} = Ax$, $y = Cx$

output: projection of the true state x onto the rows of C
 $\Rightarrow$ if C inv, then $x = C^{-1}y$

if not, then: take derivatives of $y(t)$

$$\left. \begin{aligned} \dot{y} &= C \frac{dx}{dt} = CAx \\ \ddot{y} &= CA \frac{dx}{dt} = CA^2x \\ &\vdots \\ y^{(n-1)} &= CA^{n-1}x \end{aligned} \right\} \rightarrow \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \\ \vdots \\ y^{(n-1)} \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} x := W_0 x$$

claim: if W_0 is full-rank, then we can reconstruct x

Observability

Observability

Theorem:

The system $\Sigma := (A, B, C, D)$ is observable if and only if the observability gramian W_o has full rank.

Observable Canonical Form

$$\dot{z} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} z + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0 \quad \cdots \quad 0] z + Du$$

Observable Canonical Form

$$W_o = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -a_1 & 1 & 0 & \cdots & 0 \\ -a_1^2 - a_2 & -a_1 & 1 & & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ * & * & & \cdots & 1 \end{bmatrix}$$

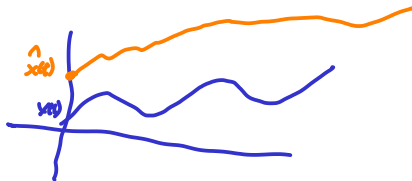
$$W_o^{-1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a_1 & 1 & 0 & \cdots & 0 \\ a_2 & a_1 & 1 & & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \cdots & 1 \end{bmatrix}$$

Observers & State Estimation

Answer to Challenge 2:

- ▶ Design a dynamical system for $\dot{\hat{x}}$
- ▶ Make $\hat{x}(t) \rightarrow x(t)$ as $t \rightarrow \infty$

Observers & State Estimation



Make a copy of the system and propagate the estimate with the right input

Original System

$$\dot{x} = Ax + Bu$$

Copy of System

$$\dot{\hat{x}} = A\hat{x} + Bu$$

Observers & State Estimation

Make a copy of the system and propagate the estimate with the right input

Original System

$$\dot{x} = Ax + Bu$$

Copy of System

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

measurement

error: $\rightarrow e$

prediction

Observers & State Estimation: Error Dynamics

$$e(t) = x(t) - \hat{x}(t) \rightarrow \text{crassidis use } e = \hat{x}(t) - x(t)$$

$$\dot{x} = Ax + Bu \quad \dot{e} = \dot{x} - \dot{\hat{x}}$$

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - \hat{y}) = Ax + Bu - (A\hat{x} + B\hat{u})$$

$$y = Cx \quad \text{later} \quad = A(x - \hat{x}) = Ae. \text{ if } A \text{ has RHP zeros,}$$

$\hookrightarrow \rightarrow 0 \text{ as } t \rightarrow \infty$

Observers & State Estimation: Error Dynamics

$$\dot{\hat{x}} = A\hat{x} + Bu + \underbrace{L(y - C\hat{x})}_{\text{gain proportional to difference between our prediction and measurement}}$$

$$\dot{e} = \dot{x} - \dot{\hat{x}} = Ax + \cancel{Bu} - (A\hat{x} + \cancel{Bu} + L(y - C\hat{x}))$$

$\hookrightarrow y = Cx$

$$= Ax - A\hat{x} - L(Cx - C\hat{x})$$

$$= A(\underbrace{x - \hat{x}}_e) - L C(\underbrace{x - \hat{x}}_e)$$

$$= (A - LC)e$$

$\hookrightarrow$ select an L so that the CL eigenvals in RHP

Duality

Controller:

$$A - BK$$

$$A \leftrightarrow A^T$$

$$B \leftrightarrow C^T$$

Observer:

$$A - LC$$

$$L \leftrightarrow L^T$$

$$W_r \leftrightarrow W_o^T$$

Duality

Controller:

$$A - BK$$

$$A \leftrightarrow A^T$$

$$B \leftrightarrow C^T$$

Observer:

$$A - LC$$

$$L \leftrightarrow L^T$$

$$W_r \leftrightarrow W_o^T$$

(A, B) reachable iff (A^T, B^T) observable; (A, C) observable iff (A^T, C^T) reachable

Observer Design by Eigenvalue Assignment

$$\frac{dx}{dt} = Ax + Bu$$

$$y = Cx$$

$$\frac{d\hat{x}}{dt} = A\hat{x} + Bu + L(y - C\hat{x})$$

$$\lambda(s) = s^n + a_1s^{n-1} + \cdots + a_n$$

Observer Design by Eigenvalue Assignment

$$\frac{dx}{dt} = Ax + Bu$$

$$y = Cx$$

$$\frac{d\hat{x}}{dt} = A\hat{x} + Bu + L(y - C\hat{x})$$

$$\lambda(s) = s^n + a_1s^{n-1} + \dots + a_n$$

$$L = W_o^{-1} \tilde{W}_o \begin{bmatrix} p_1 - a_1 \\ \dots \\ p_n - a_n \end{bmatrix}$$

$$\tilde{W}_o = \begin{bmatrix} 1 & 0 & \dots & 0 \\ a_1 & 1 & \ddots & 0 \\ a_2 & a_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-2} & a_{n-3} & \dots & 0 \\ a_{n-1} & a_{n-2} & \dots & 1 \end{bmatrix}^{-1}$$

Observer Design by Eigenvalue Assignment

$$\frac{dx}{dt} = Ax + Bu$$

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$$p(s) = s^n + p_1s^{n-1} + \dots + p_n$$

Challenge 3:

How does the controller work with estimated states?

Control with Estimated State

$y = x$, $x = c^T y$, then use that. Otherwise, ...

$u = -k \hat{x} + k_r r \rightarrow$ from LQR theory, this is optimal!

- 1) system dynamics
 - 2) observer dynamics
- feedback!

$$\dot{e} = (A - Lc)e$$

$$\hat{x} = x - e$$

$$\dot{x} = Ax + Bu$$

$$= Ax + B(-k \hat{x} + k_r r)$$

$$= Ax - Bk(x - e) + Bk_r r$$

$$= \underline{(A - Bk)}x + \underline{Bke} + Bk_r r$$

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - Bk & Bk \\ 0 & A - Lc \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix} + \begin{bmatrix} Bk_r \\ 0 \end{bmatrix} r$$

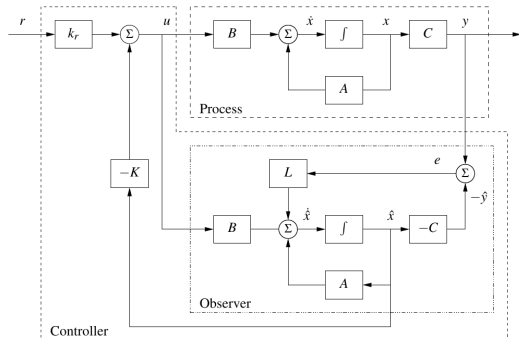
$\bar{A} \rightarrow$ b/n diagonal

$$\det(sI - \bar{A}) = \det(sI - (A - Bk)) \times \det(sI - (A - Lc))$$

Control with Estimated State

Separation Principle

The controller and observer gains can be designed **independently** of each other



$$\frac{d}{dt} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ \tilde{x} \end{bmatrix} + \begin{bmatrix} Bk_r \\ 0 \end{bmatrix} r$$

$$\lambda(s) = \det(sI - A + BK) \det(sI - A + LC)$$

Kalman Decomposition

Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & 0 & 0 \\ 0 & A_{r\bar{o}} & 0 & 0 \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & 0 & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$

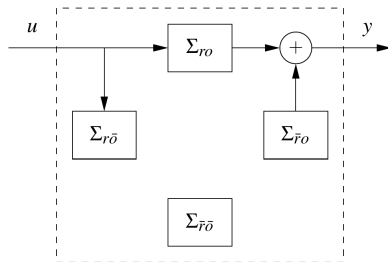
Non-Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & * & 0 \\ * & A_{r\bar{o}} & * & * \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & * & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$
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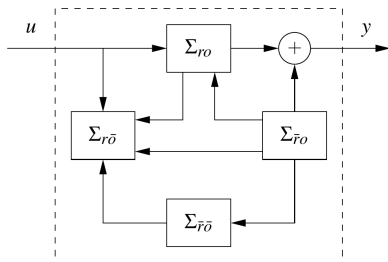
Kalman Decomposition

Distinct Eigenvalues

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$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$



Kalman Decomposition



Non-Distinct Eigenvalues

$$\frac{dx}{dt} = \begin{bmatrix} A_{ro} & 0 & * & 0 \\ * & A_{r\bar{o}} & * & * \\ 0 & 0 & A_{\bar{r}o} & 0 \\ 0 & 0 & * & A_{\bar{r}\bar{o}} \end{bmatrix} x + \begin{bmatrix} B_{ro} \\ B_{r\bar{o}} \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} C_{ro} & 0 & C_{\bar{r}o} & 0 \end{bmatrix} x + Du$$

Back to Challenge 2:

We saw optimal gains for control, what about observers?

How to choose the observer gain optimally?

In general, measurements are **noisy**.

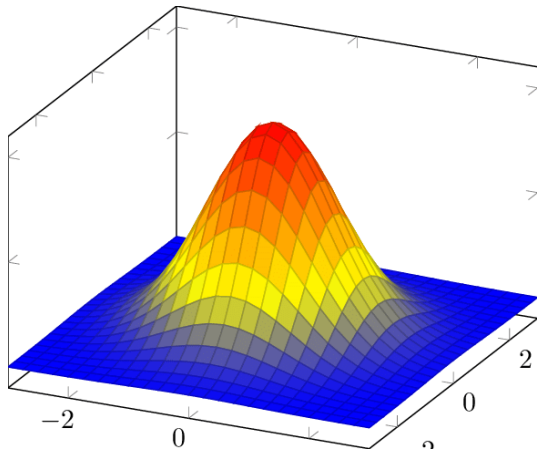
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$

$$y_k = Cx_k + Du_k + w_k$$

$$E[v_k] = 0, \quad E[w_k] = 0$$

$$E[v_k v_{k+j}^T] = \begin{cases} 0 & k \neq j \\ Q_k & k = j \end{cases}$$

$$E[w_k w_{k+j}^T] = \begin{cases} 0 & k \neq j \\ R_k & k = j \end{cases}$$



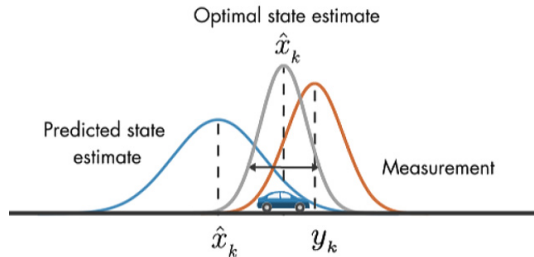
How to choose the observer gain?

$$\hat{\tilde{x}}(t) = Ax + Bu + L(\tilde{y} - Cx) \quad \rightarrow \tilde{y} = y + v(t)$$

With noisy measurements, the estimate is now a **random variable**.

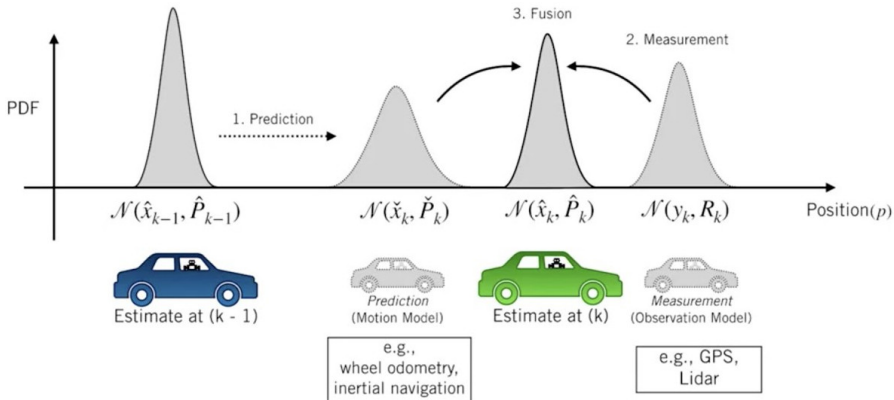
If L stabilizes the observer dynamics, then

$$E[e_k] = E[\hat{x}_k - x_k] \rightarrow 0.$$



Instead, we can optimize the 'width' (or, covariance) of the estimate

Optimal Observers



Kalman Filter Derivation

$$x_{k+1} = Ax_k + Bu_k + Fw_k$$

$$\tilde{y}_k = Cx_k + v_k$$

$$e_k = \hat{x}_k - x_k$$

(Crassidis notation,
doesn't matter!)

$$\mathbb{E}[v_k w_k^T] = 0, \quad \mathbb{E}[w_k w_j^T] = \begin{cases} 0 & k \neq j \\ Q_k & k = j \end{cases}, \quad \mathbb{E}[v_k v_j^T] = \begin{cases} 0 & k \neq j \\ R_k & k = j \end{cases}$$

at each timestep k , 2 things happen

propagation: $\hat{x}_{k+1}^- = A\hat{x}_k^+ + Bu_k$

$+$: updated

$-$: not updated

update: $\hat{x}_k^+ = \hat{x}_k^- + K_k(\tilde{y}_k - C\hat{x}_k^-)$

error covariance $P_k^+ := \mathbb{E}[e_k^+ e_k^{+T}]$

$$e_k^+ := \hat{x}_k^+ - x_k$$

Kalman Filter Derivation

First: get P_{k+1}^- as fcn of P_k^+

$$e_{k+1}^- = Ae_k^+ - Fw_k$$

(from update step)

$$P_{k+1}^- = \mathbb{E}[e_{k+1}^- e_{k+1}^{-T}]$$

$$= \mathbb{E}[(Ae_k^+ - Fw_k)(Ae_k^+ - Fw_k)^T]$$

e_k^+ depends on w_{k-1}

$$\mathbb{E}[w_k w_{k-1}^T] = 0$$

$$= \mathbb{E}[Ae_k^+ e_k^{+T} A^T] + \mathbb{E}[Fw_k w_k^T F^T] - \mathbb{E}[Fw_k e_k^{+T} A^T] - \mathbb{E}[Ae_k^+ w_k^T F^T]$$

uncorrelated, $= 0$

$$= A \underbrace{\mathbb{E}[e_k^+ e_k^{+T}]}_{P_k^+} A^T + F \underbrace{\mathbb{E}[w_k w_k^T]}_{Q_k} F^T$$

$$\Rightarrow P_{k+1}^- = A P_k^+ A^T + F Q_k F^T$$

Kalman Filter Derivation

Want: P_k^+ in terms of K_k , so we can optimize it

$$\hat{x}_k^+ = \hat{x}_k^- + K_k (\hat{y}_k - C \hat{x}_k^-)$$

$$= \hat{x}_k^- + K_k (C x_k + v_k - C \hat{x}_k^-)$$

$$= (I - K_k C) \hat{x}_k^- + K_k (C x_k + k_k v_k) \rightarrow \text{plug this into } e_k^+$$

$$e_k^+ = \hat{x}_k^+ - x_k$$

$$= (I - K_k C) \hat{x}_k^- - x_k + K_k (C x_k + k_k v_k)$$

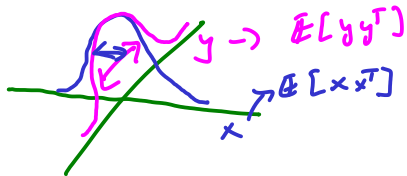
$$= (I - K_k C) \hat{x}_k^- - (I - K_k C) x_k + K_k v_k$$

$$= (I - K_k C) (\underbrace{\hat{x}_k^- - x_k}_{e_k^-}) + K_k v_k$$

Kalman Filter Derivation

$$\begin{aligned}
 P_n^+ &= \mathbb{E}[e_n^+ e_n^{+T}] \\
 &= \mathbb{E}[(I - k_n c) e_n^- e_n^{-T} (I - k_n c)^T] + \mathbb{E}[k_n v_n v_n^T k_n^T] + \text{cross terms} \\
 &= (I - k_n c) \underbrace{\mathbb{E}[e_n^- e_n^{-T}]}_{P_n^-} (I - k_n c) + k_n \underbrace{\mathbb{E}[v_n v_n^T]}_{R_n} k_n^T
 \end{aligned}$$

$= 0$
 uncorrelated noise



"minimize error covariance"
 - min the "widths" in each direction

$$\sum_{i=1}^n \mathbb{E}[(e_n^+)_i (e_n^+)_i] = \text{Tr}(P_n^+)$$

$(P_n^+)_{ii}$

Kalman Filter Derivation

$$P_n^+ = (I - K_n C) P_n^- (I - K_n C) + K_n R K_n$$

min $\text{tr}(P_n^+)$ wrt K_n

$$\frac{\partial}{\partial K_n} \text{Tr}(P_n^+) = 0 = -2(I - K_n C) P_n^- C^T + 2 K_n R$$

$$K_n (C P_n^- C^T + R) = P_n^- C^T$$

$$K_n = P_n^- C^T (C P_n^- C^T + R)^{-1}$$

Kalman Filter Derivation

Kalman Filter Derivation

Kalman Filter

Model:

$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Kalman Filter

Model:
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Initialize: $\hat{x}_0 = x_0, P_0 = P$

Kalman Filter

Model:
$$\begin{aligned}x_{k+1} &= Ax_k + Bu_k + Fv_k \\y_k &= Cx_k + Du_k + w_k\end{aligned}$$

Initialize: $\hat{x}_0 = x_0, P_0 = P$

Gain:
$$K_k = P_k^- C^T (CP_k^- C^T + R_k)^{-1}$$

Kalman Filter

Model:
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Initialize:
$$\hat{x}_0 = x_0, P_0 = P$$

Gain:
$$K_k = P_k^- C^T (CP_k^- C^T + R_k)^{-1}$$

Update:
$$\hat{x}_k^+ = \hat{x}_k^- + K_k (\tilde{y}_k - C\hat{x}_k^-)$$
$$P_k^+ = (I - K_k C) P_k^-$$

Kalman Filter

Model:
$$x_{k+1} = Ax_k + Bu_k + Fv_k$$
$$y_k = Cx_k + Du_k + w_k$$

Initialize: $\hat{x}_0 = x_0, P_0 = P$

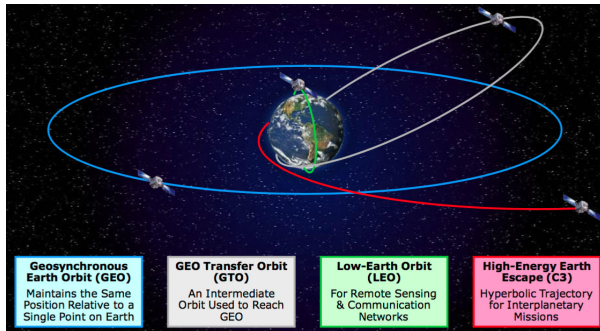
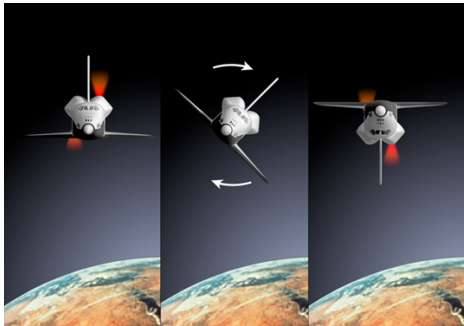
Gain:
$$K_k = P_k^- C^T (CP_k^- C^T + R_k)^{-1}$$

Update:
$$\hat{x}_k^+ = \hat{x}_k^- + K_k (\tilde{y}_k - C\hat{x}_k^-)$$
$$P_k^+ = (I - K_k C) P_k^-$$

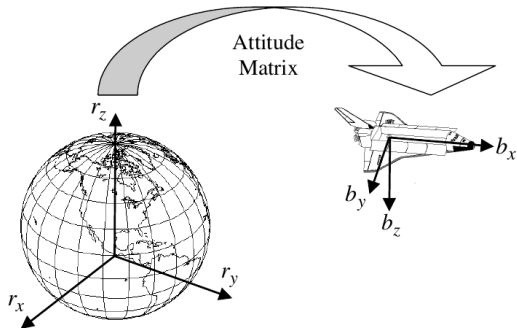
Propagate:
$$\hat{x}_{k+1}^- = A\hat{x}_k^+ + Bu_k$$
$$P_{k+1}^- = AP_k^+ A^T + BQ_k B^T$$

Spacecraft GNC

Spacecraft GNC



Attitude



Reference and body frames:

$$b = b_x \hat{b}_1 + b_y \hat{b}_2 + b_z \hat{b}_3$$

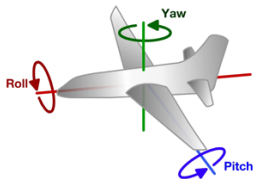
$$r = r_x \hat{r}_1 + r_y \hat{r}_2 + r_z \hat{r}_3$$

Transformation between the two:

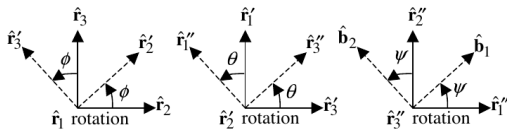
$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = A \begin{bmatrix} \hat{r}_1 \\ \hat{r}_2 \\ \hat{r}_3 \end{bmatrix} \quad (1)$$

Euler Angles

One can represent a rotation via the **Euler angles**: roll ϕ , pitch θ , and yaw ψ .

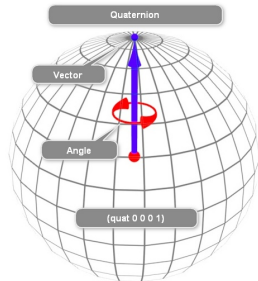


$$r' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} r \quad r'' = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} r' \quad b = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} r''$$



Quaternions

The Euler angles have a singularity when the attitude is pointed at a pole, so spacecraft attitude is represented as a **quaternion** instead.



$$q = \begin{bmatrix} \rho \\ q_4 \end{bmatrix} \in \mathbb{R}^4$$

$$q^T q = 1$$

$$A(q) = \Xi^T(q)\Psi(q)$$

$$\Xi(q) = \begin{bmatrix} q_4 I_{3 \times 3} + [\rho \times] \\ -\rho^T \end{bmatrix}$$

$$\Psi(q) = \begin{bmatrix} q_4 I_{3 \times 3} - [\rho \times] \\ -\rho^T \end{bmatrix}$$

$$[\rho \times] = \begin{bmatrix} 0 & -\rho_3 & \rho_2 \\ \rho_3 & 0 & -\rho_1 \\ -\rho_2 & \rho_1 & 0 \end{bmatrix}$$

Rigid Body Dynamics

Dynamics: Euler Angles

Kinematics: Quaternions

► H : Angular momentum

$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega \times]J\omega + L\end{aligned}$$

$$\begin{aligned}\dot{q} &= \frac{1}{2}\Omega(\omega)q \\ \Omega(\omega) &= \begin{bmatrix} -[\omega \times] & \omega \\ -\omega^T & 0 \end{bmatrix}\end{aligned}$$

The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

Rigid Body Dynamics

Dynamics: Euler Angles

► H : Angular momentum

► J : Moment of Inertia

$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega \times]J\omega + L\end{aligned}$$

Kinematics: Quaternions

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Rigid Body Dynamics

Dynamics: Euler Angles

- ▶ H : Angular momentum
- ▶ J : Moment of Inertia
- ▶ L : Applied torque

$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega \times]J\omega + L\end{aligned}$$

Kinematics: Quaternions

$$\begin{aligned}\dot{q} &= \frac{1}{2}\Omega(\omega)q \\ \Omega(\omega) &= \begin{bmatrix} -[\omega \times] & \omega \\ -\omega^T & 0 \end{bmatrix}\end{aligned}$$

The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

Rigid Body Dynamics

Dynamics: Euler Angles

- ▶ H : Angular momentum
- ▶ J : Moment of Inertia
- ▶ L : Applied torque
- ▶ ω : Angular velocity

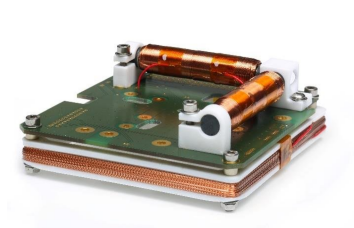
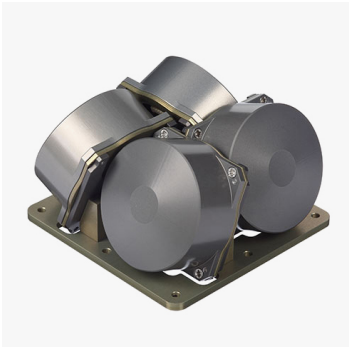
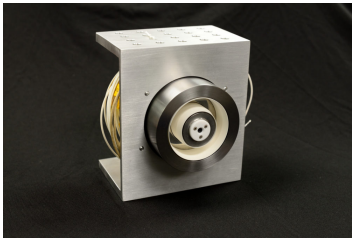
$$\begin{aligned}\dot{H} &= J\dot{\omega} \\ &= -[\omega \times]J\omega + L\end{aligned}$$

Kinematics: Quaternions

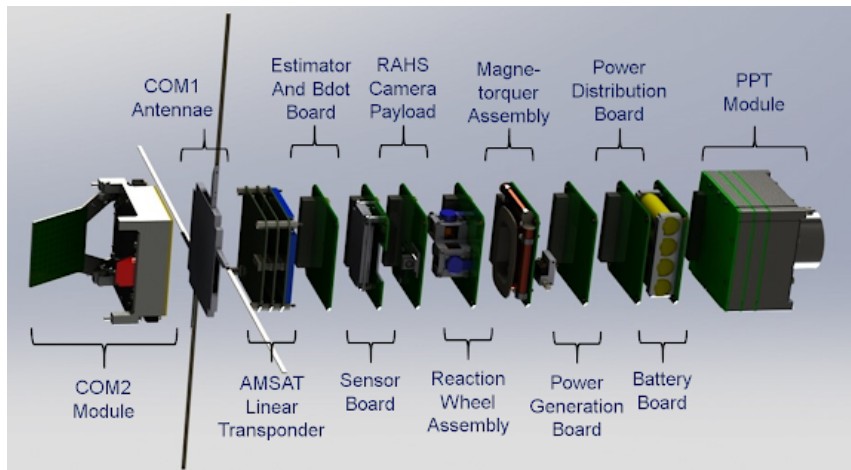
$$\begin{aligned}\dot{q} &= \frac{1}{2}\Omega(\omega)q \\ \Omega(\omega) &= \begin{bmatrix} -[\omega \times] & \omega \\ -\omega^T & 0 \end{bmatrix}\end{aligned}$$

The torque L comes from **thrusters**, **reaction wheels** or **magnetorquers**.

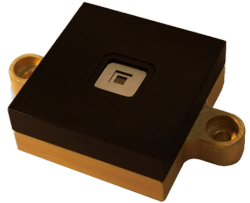
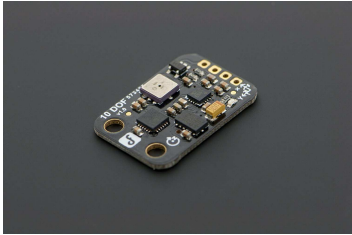
Spacecraft Actuators



Plasma Thruster

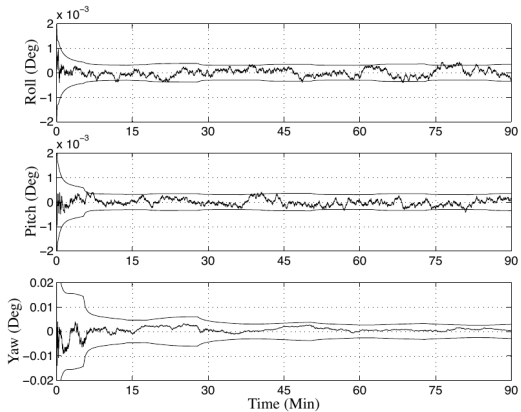


Measurement



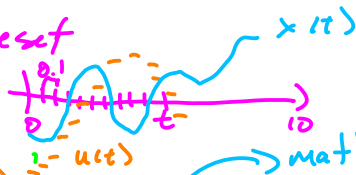
Quaternion Extended Kalman Filter

Initialize	$\hat{\mathbf{q}}(t_0) = \hat{\mathbf{q}}_0, \quad \hat{\beta}(t_0) = \hat{\beta}_0$ $P(t_0) = P_0$
Gain	$K_k = P_k^- H_k^T (\hat{\mathbf{s}}_k^-) [H_k(\hat{\mathbf{s}}_k^-) P_k^- H_k^T (\hat{\mathbf{s}}_k^-) + R]^{-1}$ $H_k(\hat{\mathbf{s}}_k^-) = \begin{bmatrix} A(\hat{\mathbf{q}}^-) \mathbf{r}_1 \times & 0_{3 \times 3} \\ \vdots & \vdots \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_n \times & 0_{3 \times 3} \end{bmatrix} \Big _{l_k}$
Update	$P_k^+ = [I - K_k H_k(\hat{\mathbf{s}}_k^-)] P_k^-$ $\Delta \hat{\mathbf{s}}_k^+ = K_k [\bar{\mathbf{y}}_k - \mathbf{h}_k(\hat{\mathbf{s}}_k^-)]$ $\Delta \hat{\mathbf{s}}_k^+ \equiv [\delta \hat{\alpha}_k^{+T} \quad \Delta \hat{\beta}_k^{+T}]^T$ $\mathbf{h}_k(\hat{\mathbf{s}}_k^-) = \begin{bmatrix} A(\hat{\mathbf{q}}^-) \mathbf{r}_1 \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_2 \\ \vdots \\ A(\hat{\mathbf{q}}^-) \mathbf{r}_n \end{bmatrix} \Big _{l_k}$ $\hat{\mathbf{q}}_k^+ = \hat{\mathbf{q}}_k^- + \frac{1}{2} \Xi(\hat{\mathbf{q}}_k^-) \delta \hat{\alpha}_k^+, \quad \text{re-normalize quaternion}$ $\hat{\beta}_k^+ = \hat{\beta}_k^- + \Delta \hat{\beta}_k^+$
Propagation	$\dot{\omega}(t) = \dot{\omega}(t) - \hat{\beta}(t)$ $\dot{\hat{\mathbf{q}}}(t) = \frac{1}{2} \Xi(\hat{\mathbf{q}}(t)) \dot{\omega}(t)$ $\dot{P}(t) = F(t) P(t) + P(t) F^T(t) + G(t) Q(t) G^T(t)$ $F(t) = \begin{bmatrix} -[\dot{\omega}(t) \times] & -I_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}, \quad G(t) = \begin{bmatrix} -I_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix}$



Matlab - simulating ODEs

```
1 clear all; close all; clc;
2
3 % simulation timespan
4 dt=0.1;
5 t_space = [0:dt:10];
6
7 % initial condition
8 x_0 = [10;10];
9
10 % sinusoidal input with zero-order-hold
11 u = cos(t_space);
12 x_array = zeros(2, length(t_space));
13
14 x_array(:,1) = x_0;
15
16 for ii = 1:length(t_space)-1
17     t_int = [t_space(ii), t_space(ii+1)];
18
19     %simulate ODE with input
20     [t_out,y_out] = ode45(@fn_rhs,t_int,x_array(:,ii),[],u(ii));
21     x_array(:,ii+1) = y_out(end,:);
22
23 end
24
25 plot(t_space, x_array(1,:)), hold on
26 plot(t_space, x_array(2,:))
```



```
1 function [x_dot] = fn_rhs(t, x, u)
2 %fn_rhs: right-hand-side function of the ode x_dot = f(t,x,u)
3
4 x_dot = exp(-t)*[cos(x(1)); sin(x(2))] + exp(-0.1*t)*[0;1]*u;
5
6 end
```

$\dot{x} = f(x, t, u) \rightarrow$ separate function

ODE45
option
↑
rest of inputs
↓
initial condition

matlab will tell you if
ODE45 is wrong



